

ICEIS 2026 - Paper #143

Paper Title: *LLM-Driven Ontology Learning: From Term Extraction to Upper-Level Categorization for building Information System models*

Reviewer #1

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 4
Originality (Newness of the ideas expressed): 2
Technical Quality (Theoretical soundness/methodology): 2
Significance (Is the problem worth the given attention?): 2
Presentation (Structure/Length/English): 3
Overall Rating (Weighted value of above items): 3

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes
Needs more experimental results? Yes
Needs comparative evaluation? Yes
Improve critical discussion ? (validation): Yes
Figures are adequate ? (in number and quality): Yes
Conclusions/Future Work are convincing? No
References are up-to-date and appropriate? Yes
Paper formatting needs adjustment? No
Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

The paper discusses the pressing issue of the bottleneck of knowledge acquisition in Ontology engineering. The authors present their idea to force LLMs to generate a natural language term definition before categorizing the term into hierarchies.

Overall, the paper fails to prove that their method can help the ontology engineering process as no actual ontology was generated in the end. This makes the results of the paper difficult to compare with existing LLM-based ontology generation methods mentioned in the Background section. Furthermore, the manuscript does not provide a clear justification for why these methods were not included as a baseline for comparison.

In the following my detailed feedback:

1. If an LLM already has the knowledge to use a definition without further context thanks to its training data, I don't understand what advantage there is in writing this out, except for explainability or chain-of-thought prompting. In my opinion, the experiments also do not prove what is claimed, as the different results in phase 2 mean that the evaluations of the subsequent phases are no longer comparable (because the input is not the same).

2. In some places, the methodology does not make it clear what exactly was done. For example, for pipeline 1 in phase 2, how does the LLM read 40 articles? All at once or in chunks one after the other?
3. Why is the temperature increased in pipeline 2 in phase 2? Why is it necessary for the LLM to become more creative here? This should be explained, and how is "factual relevance" ensured?
4. What is a category in terms of ontologies? At some other part the authors call it concepts. These inconsistencies make it very difficult to follow.
5. Phase 4: Does it really make sense to take the 100 terms from the experts as ground truth, and is it fair to evaluate Pipeline 1 of Phase 2 afterwards, since the LLM in the form of the articles may not have been provided with all of these terms and is only allowed to use them?
6. In Section 5.2, the authors claim that 91% compared to 88% would be significantly higher without proving how significant this is. Given the limited data available, I dare to doubt this without error calculation and p-value.
7. In general, the insight that LLMs need context to be able to make factual statements about terms is very trivial and almost obvious when looking at current LLM research.
8. It is not clear to me to what extent this is a scientific contribution. The authors even cite state-of-the-art work on ontology generation with LLMs and then present something that does not even generate entire ontologies, but only a taxonomy or hierarchy.

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Reviewer #2

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 6
Originality (Newness of the ideas expressed): 3
Technical Quality (Theoretical soundness/methodology): 4
Significance (Is the problem worth the given attention?): 4
Presentation (Structure/Length/English): 4
Overall Rating (Weighted value of above items): 4

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes
Needs more experimental results? Yes
Needs comparative evaluation? No
Improve critical discussion ? (validation): Yes
Figures are adequate ? (in number and quality): Yes
Conclusions/Future Work are convincing? Yes
References are up-to-date and appropriate? Yes
Paper formatting needs adjustment? No
Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

The authors present an empirical framework for ontology learning integrating four term-extraction pipelines (LLM + Corpus, LLM-Generated, NER + Corpus, and TF-IDF) with a classification step based on the production and exploitation of Natural Language Definitions for the extracted terms. The experiments being conducted aim at extracting domain relevant concepts and placing them in a taxonomy that is bootstrapped with a top ontology (BFO) and two core ontologies (combo of GeoCore and GeoReservoir). The outcome of the experimentation show how LLM-based methods outperform traditional statistical methods by leveraging NLDs. The study is applied to the Brazilian Pre-Salt geology domain and includes a blind expert evaluation of term relevance, definition accuracy, and taxonomic correctness.

The study paper is well written and an interesting reading. It contains some conceptual issues in the experimentation (see my comments below) that undermine the intended objective of presenting a solid comparison of different methods for ontology learning but has indeed some value and knowledge to offer.

SPECIFIC REMARKS

In the abstract it is not clear whether, as said in a first statement, LLMs are leveraged to produce NLDs or,

as expressed in a second statement, NLDs are leveraged to produce better term extraction and taxonomy categorization. The paper then clarifies (only in the second paragraph of section 3, as the first paragraph of the same section is still quite ambiguous with that "provided with a high-quality" NLD, but the abstract should be, yet in its simplicity, unambiguous.

In the introduction: "automate or reduce human intervention". Please rephrase: one automates a process, not the human intervention in it. So it could be something like "we aim to completely automate the steps <list of steps> or at to at least reduce human intervention in them".

This is not a negative comment on the paper, as it doesn't differentiate from lot of literature in the field; however, I find it important to address the following aspect as it represents a pivot aspect of the presented work: cited works on top/core ontologies are old, initial works on defining what an ontology is, how it is modeled, etc.. The cited Neon paper sits on top of those papers and defines a methodology. However, none of the above do scientifically support or justify the need for top/core ontologies. The sentence "To ensure semantic interoperability and future expansion, these vocabularies are defined within a hierarchical structure" does not reflect any need for top-level ontologies. An ontology can only be a discretization of reality; the Semantic Web deals with pluralism, not a single all-encompassing ontology. There is no proof that a top ontology helps in modeling (and it's an hard task measuring it) and there is not, especially when explicit links to it are asserted, a thorough study showing that its supposed advantages are not balancing the risks of potential semantic conflicts (because there is not a single top ontology). There are concrete examples of metaontologies (not top), such as OntoClean (Guarino and Welty), that provide formal rules for spotting issues in modeling, but they 1) suffer from high costs of annotation 2) are in any case a very different thing. I can only provide personal experience, i.e. I know only very few cases of real use of top ontologies if we do not consider those ontologies developed by people who developed also the top ones or those developed by non-experts that have been advised by those developing top ontologies; the most relevant ontologies (e.g. those from W3C) do not link to top ontologies and, more importantly, have not been developed thanks to the connection to top ontologies. This is anecdotal and personal, but I am missing the literature that really shows their value (and the reported one does not) as the burden of proof is on the claimant.

Now, concretely addressing the paper: if the top/core ontologies are beneficial to the creation of the ontology by the LLM and not taken as an assumption, that is a different claim that is important to be specifically clarified as a characteristic of this work.

I would suggest to formally clarify the intuitive levels of accuracy that have been listed in section 4.4.

A few questions came to my mind; just mentioning two here:

- * What is the difference between "Irrelevant" and "Invalid" when a term is simply, neutrally generated? Invalid is not even a term? (e.g. a wrong n-gram that is not a term)
- * "Correct but Unspecific" means that, given a concept C generated by the extraction and added to the taxonomy under a super concept S, there exists in the ontology another concept SS, subconcept of S, that should have been elected as superconcept of C?

Some formal definition would help and is important in the context of the presented work

Can the authors elaborate more on the setup for the TF-IDF approach? TF-IDF is a well know measure, yet it can be implemented over different ways to first extract the candidate terms (e.g. being limited to N-grams for a certain "N", be vocabulary based to filter out inexistent terms, especially on n-grams for $N > 1$, using any NLP technique for guaranteeing meaningful terms being extracted instead of merely checking windows of n tokens, etc..)

I am not sure I interpreted Table 4 correctly. I was surprised by LLM Generated being better than LLM + Corpus. Very much in general, in ML, it is sometimes possible that a supposedly good addition (the corpus) to any ML method "confuses" the system. However, considering the heavy weight of the trained LLM and the small, but potentially useful, contribution of the corpus, I would expect much more probably a (more or less effective) positive contribution, definitely not a negative one. However, the comment about the extracted terms made at the start of section 5.3 (and, indeed, the fact that an identical process has been run for the NLD generation) me realize that the terms might not be the same. So, for instance, the LLM+

Corpus has extracted better terms (tables 1–3, where table 1 especially suggests that, in a hypothetical ranking given by the choice of most relevant terms produced by humans, LLM+Corpus scored much better than LLM Generated) but, for the reason explained in section 5.3, these are harder to describe. This however – if I got it right – makes this test almost useless, as it seems to be comparing apples and oranges. The expression "to process all candidate terms from all four pipelines" in section 4.3 made me think (I wasn't sure indeed) that, at least for testing this aspect specifically, all terms produced by all methods were put together and then evaluated by all methods, while results and the sentence in 5.3 (and, again, the fact that a single workflow was devised specifically for NLD generation) suggest that each pipeline has been evaluated wrt its own term base. If I'm right, I understand the difficulties of a fair assessment even on the total number of terms (the methods based *only* on corpus cannot hope to provide info for terms that are not in the corpus), but a selection could be made, yet the terms should be the same. Without need of testing all four sets of produced terms, a possibility could have been to test the behavior of LLM+Corpus on its own produced terms (or on all terms, in order to have a common testbed), instead of using a generic LLM to produce the NLDs, thus comparing the Generic LLM with the LLM+Corpus on that respect.

The link <https://anonymous.4open.science/r/PreSaltOntology-61FA> resolves into a space in the "anonymous for open science" service; however, the four folders with the 4 approaches are empty. I've also tried downloading the full repo: the directories within it are indeed not included in the download and there is no trace of the source code and prompts mentioned in section 4.2 Also, I'm not sure whether the produced ontology is `conceptual_artifact.rdf` (which indeed corresponds to the pictures in the last figures in the paper) or was meant to be another, subsequent, step of the process.

I see there's a distinction between classes and individuals. However, while I may understand that axioms are not available and thus it is, at the current stage of automatic production, a mere taxonomy, that one is clearly not even a draft of an ontology. There are several elements that are palely not classes (e.g. Cretaceous is a prehistorical period, then an individual, what can an instance of Cretaceous be?). If Cretaceous cannot be a class, then the `rdfs:subClassOf` assertion is also wrong. Indeed, I wanted to check what GEOCORE_0000005 is in order to verify my impression but, unfortunately, the URI does not resolve. I thus google it and found the geocore ontology on github (much to the dismay of LOD principles but that's not on the authors). So, side note for the authors: this is not necessary to the purpose of publishing your ontology "per se" but, if published as a shared data for your paper, given the present status of the geocore ontology, I suggest to add labels of the extended elements from Geocore as comments in the ontology so the interested reader doesn't have to go looking for Geocore on the web Back to my investigation, GEOCORE_0000005 is indeed representing "geological time interval" (in line with what shown on the right class tree of figure 3). Then I confirm that, given the intension of GEOCORE_0000005, "Cretaceous" is not a subclass of it, it's an instance.

While this might be not relevant to the scope of the paper, which is focused on taxonomy construction, why not simply adopting a different terminology in the paper? Thesauri are a thing (and there's a standard for that : SKOS). Besides any technicism, I suggest to simply speak of taxonomies, or thesauri, or domain vocabularies in general. I understand that the "culprit" possibly goes back to the so called "top ontologies", which might have been subject to the same issues, but then speaking of axiomatization is controversial. What axiomatization can there be if right from the very few axioms being used, they are wrong?

It would have been interesting to see an approach not relying on NLDs. In the end, NLDs are also a product of the first steps and while explicitly exposing a resource can improve the quality of further steps, the resource (especially in the case of LLM) is something already inherent the LLM itself

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Reviewer #3

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 6
Originality (Newness of the ideas expressed): 5
Technical Quality (Theoretical soundness/methodology): 5
Significance (Is the problem worth the given attention?): 5
Presentation (Structure/Length/English): 4
Overall Rating (Weighted value of above items): 5

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes
Needs more experimental results? No
Needs comparative evaluation? No
Improve critical discussion ? (validation): Yes
Figures are adequate ? (in number and quality): Yes
Conclusions/Future Work are convincing? Yes
References are up-to-date and appropriate? Yes
Paper formatting needs adjustment? No
Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

The paper is positioned in the rich scientific area of ontology building. In particular, it focuses on the first step that is the identification of the terminology and its classification according to upper-level meta-types.

The proposal consists in analysing and comparing different approaches: LLM + Corpus, LLM Generated, NER Model + Corpus, and TF-IDF. The results, evaluated by domain experts, show that LLM+Corpus is more effective than the other approaches.

The overall quality is good, the presentation is clear in general. There are specific passages that should be clarified. See below.

— Specific comments —

- In certain passages, the text is somewhat obscure. For instance, a clarification is needed here: Natural Language Definitions (NLDs), even across multilingual settings. Primarily, you should clearly define NLD. It is an intuitive phrase, but it is used in technical terms. Please clarify upfront.
- Pipeline 1. It does not specify anything about the Corpus.
 - purely extractive mode. Please clarify. And what's the difference wrt Pipeline 2? Is it not extractive? What is the alternative to the extractive mode?
 - What is NER? Please clarify unusual acronyms when first introduced

- Aristotelian–style definition (e.g., "X is a Y that Z"). Obscure
- Five domain specialists provided a reference set of 100 terms (20 terms each). Funny no overlap between the 5 sets of terms?

If you add somewhere (you decide where) a few examples of some terminology/ontology content, the paper would gain in quality.